phase4 system design

1. Objective of Phase 4

Upgrade system from:

network-aware credibility

to:

spatially-aware information system with localized relevance and alerts

2. What Phase 4 Adds

Core Additions

User location tracking L_i(t) Location confidence modeling Post location modeling l_j Proximity function Prox(u,j,t) Spatial filtering in alerts Spatial trust aggregation

Still NOT included

ML content models Memory similarity Advanced NLP

3. Core Idea

A post should not be shown or alerted globally just because it is credible.

It must also satisfy:

“Is this relevant to this user’s location?”

4. User Location Model

4.1 Raw Location

l_i(t) = (,)

Collected from:

- GPS (preferred)
- IP (fallback)

4.2 Location Confidence

Formula

L_i(t) = w_1 S_{} + w_2 S_{} + w_3 S_{} + w_4 S_{}

Meaning

- aggregates all location trust signals
- outputs value in [0,1]

4.3 Interpretation

High L_i → location is reliable Low L_i → location is suspicious

5. Post Location

Definition

l_j =

Collection Options

User-provided location OR location inferred from first trusted users

Fallback Strategy

if post has no location: estimate l_j = weighted avg of user locations

6. Proximity Function

Formula

Prox(u,j,t) = L_u(t) (-)

Meaning

- combines:
  - geographic distance
  - trust in location

7. Spatial Trust of Post

Formula

{L}_j(t) =

Meaning

- average location reliability of contributors

8. Updated Propagation Condition

Phase 4 Expand Rule

Expand(j,t) = (C_j ;; N_j N_{min} ;; Var_j ^2 ;; {L}*j L*{min})

9. Updated Alert Function

Phase 4 Alert

Alert(u,j,t) = (Prox(u,j,t) *p ;; C_j* {alert} ;; Var_j ^2)

10. Pipeline (Phase 4)

User action ↓ Collect location + compute L_i ↓ Update user + graph (Phase 3) ↓ Update post credibility ↓ Compute post location l_j ↓ Compute: Prox(u,j,t) $\bar{L}_j(t)$ ↓ Propagation decision ↓ User-specific alert decision

11. What Phase 4 Solves

Problem 1: Irrelevant Alerts

Global alerts for local events

Solution:

Prox(u,j,t) filters by distance

Problem 2: Location Spoofing

Fake location reports

Solution:

Low L_i → low influence

Problem 3: Regional Trust

Users far away influencing local events

Solution:

Proximity reduces their impact

12. Backend Changes

User Table Update

ALTER TABLE users ADD COLUMN: lat FLOAT, lon FLOAT, location_conf FLOAT

Post Table Update

ALTER TABLE posts ADD COLUMN: lat FLOAT, lon FLOAT, location_conf FLOAT

Location Update API

POST /location { “user_id”: “u1”, “lat”: 12.34, “lon”: 56.78 }

13. Optimization Strategy

Spatial Indexing

Use: PostGIS OR geo-hash indexing

Filtering

only compute Prox for posts within radius R

14. Hyperparameters (Phase 4)

sigma_p → spatial decay tau_p → proximity threshold L_min → location trust threshold

15. What to Test

Test 1: Local Event

Fire in one city

Expect:

Nearby users → alerted Far users → ignored

Test 2: Spoofed Location

User jumps locations

Expect:

L_i ↓ → influence ↓

Test 3: Mixed Users

Local + remote users voting

Expect:

local trusted users dominate

16. Phase 4 Summary

Phase 3: “network matters” Phase 4: “location matters”

17. System Evolution

Phase Capability MVP basic Phase 2 user-aware Phase 3 network-aware Phase 4 geo-aware

18. Final Insight

After Phase 4, your system becomes:

context-sensitive and regionally intelligent

This is what makes it usable in:

- emergency systems
- local information platforms
- real-time event detection